Chapter 4

CONCLUSIONS AND RECOMMENDATIONS

This chapter presents conclusions drawn from the findings of WorkPulse: Employee Attendance and Work Analytics Information System for MVL and recommendations for further improvement. User-evaluation results are distinguished from technical testing and biometric performance evidence.

Conclusions

1. The study identified inaccurate attendance recording, risks of unauthorized or proxy attendance, inefficient work-hour monitoring, payroll-processing difficulties, and limited employee access to records as major challenges. These findings support the need for centralized workforce information management.

2. WorkPulse provides centralized employee records for the Regular, Extra, and Manager / Supervisor employee classifications, alongside administrator and employee access groups. The documented functions include fingerprint-based attendance, authorized manual attendance and corrections with required reasons and correction history, employee notifications, leave requests, cash advance management, payroll and payslip summaries, descriptive reporting, and analytical features. Cash advance workflows encompass requests, approval or rejection, confirmation of disbursement, outstanding balances, and repayment tracking. Payroll workflows distinguish Ready to Pay, Carried to Next Period, and Paid statuses. Limited offline saving and synchronization support designated operations but do not make the entire system offline-capable.

3. The reported System Usability Scale (SUS) score was 80.83, interpreted in Chapter 3 as Good and Acceptable. This indicates favorable usability feedback from participating respondents; it does not, by itself, establish technical correctness or operational reliability.

4. The respondent questionnaire reported weighted means of 4.90 for security, 4.69 for performance efficiency, and 4.88 for reliability, with an overall weighted mean of

4.82. All were interpreted as Strongly Agree. These are respondents' perceptions of system quality, not direct measurements of security, speed, or reliability. Technical test outcomes must be reported and interpreted separately using actual expected-versus-observed results.

5. Fingerprint attendance depends on the configured scanner, its locally installed capture and matching components, and the designated workstation. The study's fingerprint-performance objective must be concluded using actual genuine and unrecognized scan trials, timing observations, and the documented computation of relevant rates. [INSERT VERIFIED BIOMETRIC RESULTS: trial counts, valid comparisons, false acceptances, false rejections, failed captures, processing times, and corresponding rates.] No numerical biometric performance conclusion is asserted without those records.

6. WorkPulse has operational limitations. Fresh authentication, server-side approvals, fingerprint transactions, and payment-related actions require their applicable online services or local workstation components. Offline queues and cached views are limited and may contain outdated records; queued actions require synchronization and server revalidation. When the fingerprint workstation or device is unavailable, the approved fallback is verified administrator manual attendance with a documented reason, correction history, and employee notification; a paper attendance record may be used until authorized entry is possible.

Recommendations

1. Conduct broader fingerprint-reader and workstation compatibility tests under different device conditions. Maintain documented workstation recovery steps, reader diagnostics, and a paper-to-authorized-manual-attendance fallback procedure for device failure.

2. Improve attendance monitoring beyond the existing daily/date-range filters and basic summaries by adding exception-focused views, correction review tools, clearer source indicators, and configurable alerts for unusual or unresolved attendance records.

3. Extend payroll and cash advance reporting with reconciliation views, repayment histories, carried-balance audit trails, and clearer employee-facing breakdowns. Verify

that settled payroll records cannot be altered by attendance corrections without an approved controlled process.

4. Enhance the existing employee notification feature through configurable notification preferences, clearer delivery status, and optional additional delivery channels, subject to consent and data-protection requirements.

5. Further evaluate and refine existing descriptive and AI-assisted analytical functions using sufficient relevant data, documented model versions and performance measures where trained models are active, and explicit labels when statistical fallback is used. Do not treat forecasts as automatic personnel or payroll decisions.

6. Conduct extended outage, synchronization, and recovery tests, including duplicate submissions, stale cached data, local-storage failures, unauthorized access, and interrupted connectivity. Record expected and observed outcomes before drawing technical reliability conclusions.

7. Complete and report biometric evaluation using actual enrollment and scan-trial evidence, including genuine and unrecognized trials, failed captures, measured response times, and clearly defined FAR/FRR denominators. Expand testing only with appropriate consent and privacy safeguards.

8. Preserve complete supporting materials, including respondent questionnaires, SUS scoring sheets, item-level Likert calculations, technical test logs, biometric evaluation records, and current deployment screenshots, as appendices or verifiable project evidence.

Evidence required before final submission

The uploaded Chapter 4 and previously shared Chapter 3 report survey scores but do not provide verified biometric trial counts or complete technical test logs. Replace the bracketed biometric placeholder and substantiate technical conclusions only after reviewing the actual evidence. Confirm that every feature described as implemented is present in the final tested deployment.